

THE SKILLS AND TRAITS OF ENTREPRENEURS

LECTURE 2

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THE INVESTMENT CHOICE

You have €100,000 to invest for 10 years. Which do you choose?

OPTION A

Start your own company

You put most of your wealth into one private firm where you also work.

- About **35%** chance the business survives the first 10 years
- About **10%** chance you lose everything you invested
- Median capital gain: **6.9%** per year
- Upper quartile: **18.6%** per year
- About **10%** chance **>200%** excess gains per year vs. the public of stock market

OPTION B

Invest in the public stock market

You buy a diversified public equity index. Over comparable horizons, public equity returns are:

- Minimum: **5.6%** per year
- Maximum: **19.9%** per year
- Median excess gains compared to entrepreneurs: **2%** per year

THE ENTREPRENEURIAL INVESTMENT PREMIUM

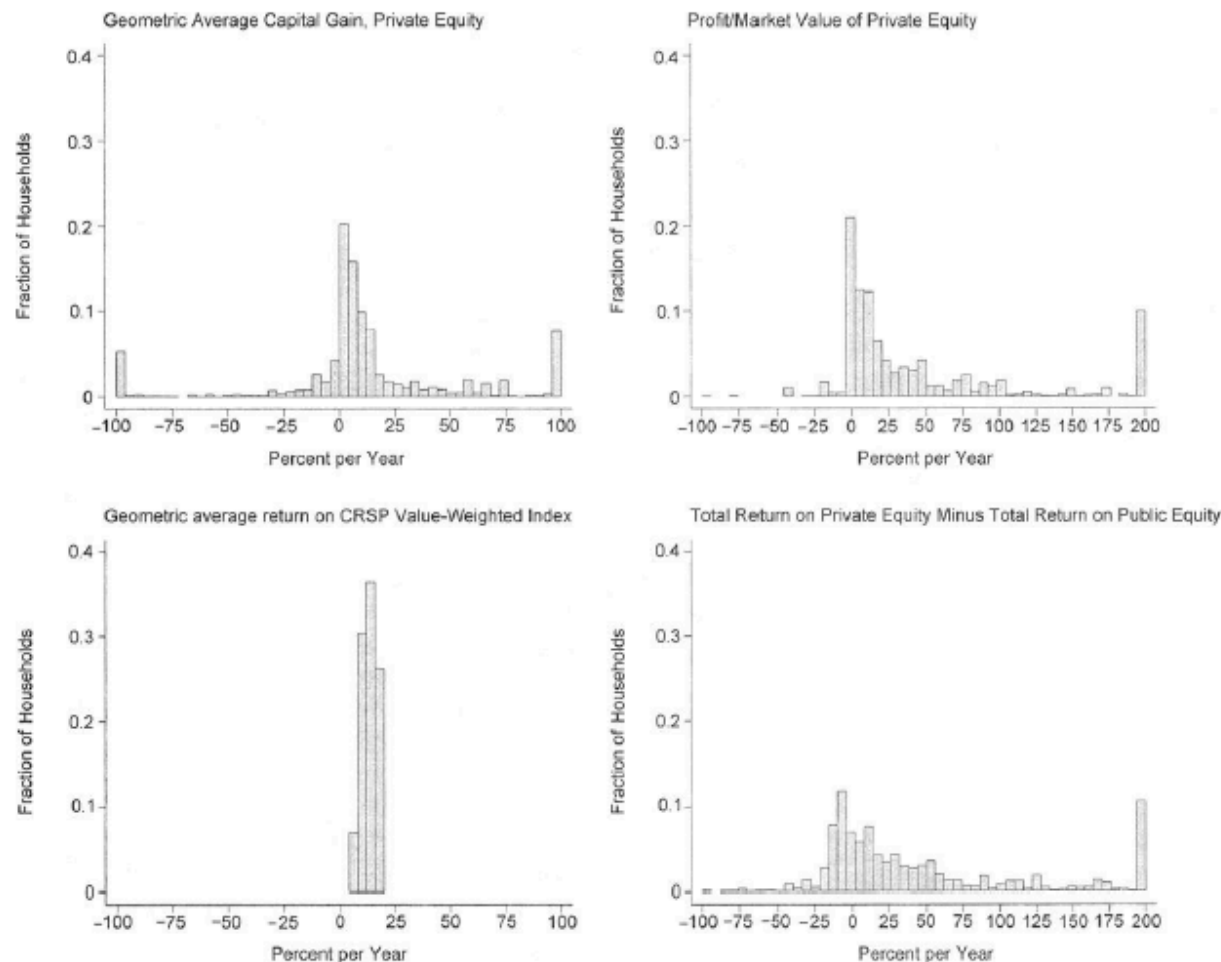


FIGURE 2. THE CONDITIONAL DISTRIBUTION OF RETURNS TO PRIVATE EQUITY ACROSS HOUSEHOLDS

Notes: Household data from the 1989 SCF are used to plot the returns to private equity investment in surviving firms. The top left plot shows the histogram of geometric average annual capital gains accrued across households. The top right plot shows the histogram of earnings rates (earnings in the year prior to the survey divided by market value of equity) accrued across households. The bottom left plot shows the histogram across households of the geometric average return on investment if households had instead invested their wealth in the CRSP value-weighted index of all publicly traded equity over the same horizon as their private equity investment. The bottom right plot shows the histogram across households of the total average excess of the CRSP index return over each household's holding period.

Source: Moskowitz, T. J. & Vissing-Jørgensen, A. (2002). The Returns to Entrepreneurial Investment: A Private Equity Premium Puzzle? *American Economic Review*, 92(4), 745-778.

THE ENTREPRENEURSHIP PUZZLE

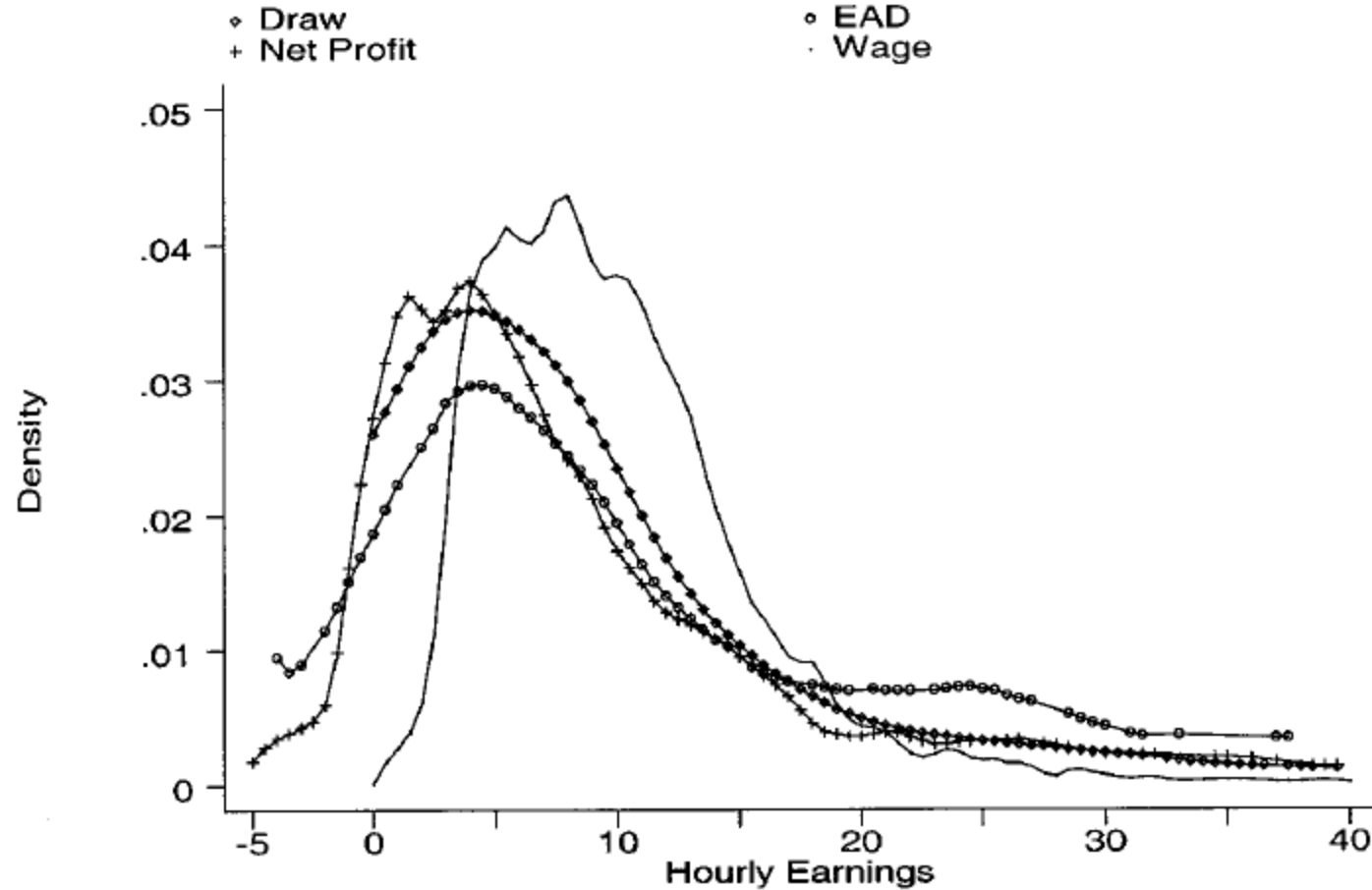


FIG. 1.—Empirical distributions, hourly earnings measures

THE ENTREPRENEURSHIP PUZZLE

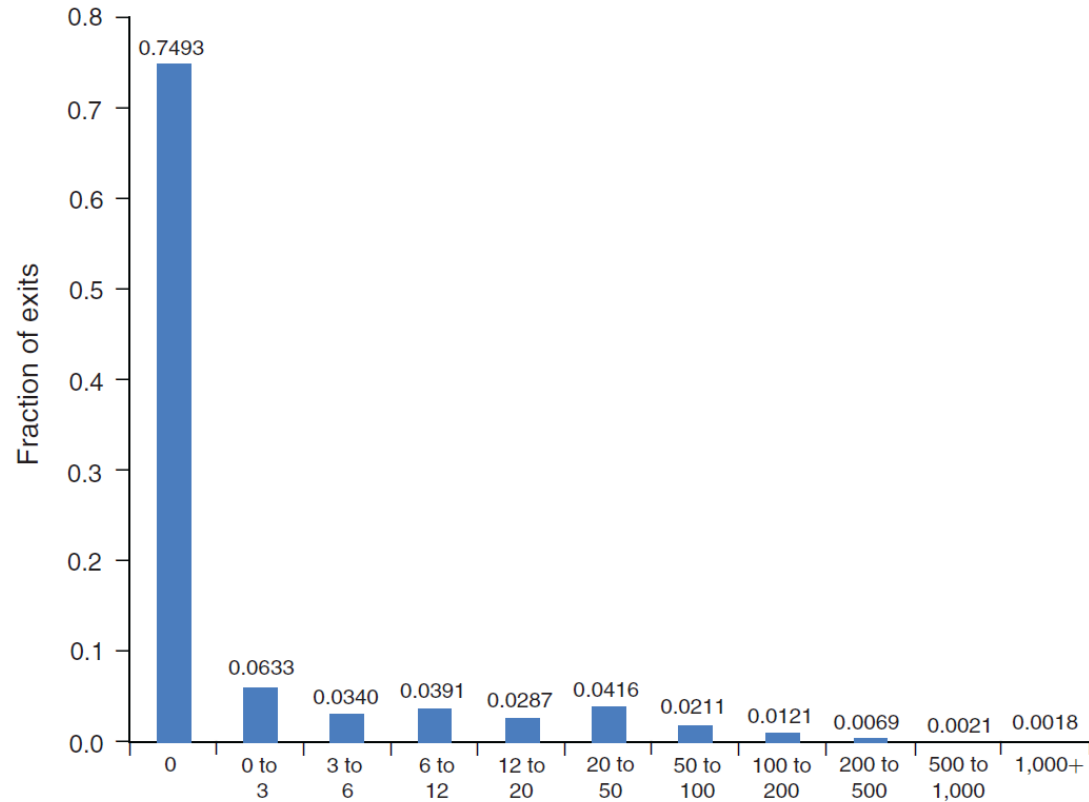


FIGURE 2. MARGINAL DISTRIBUTION OF EXIT VALUE

FIGURE 2

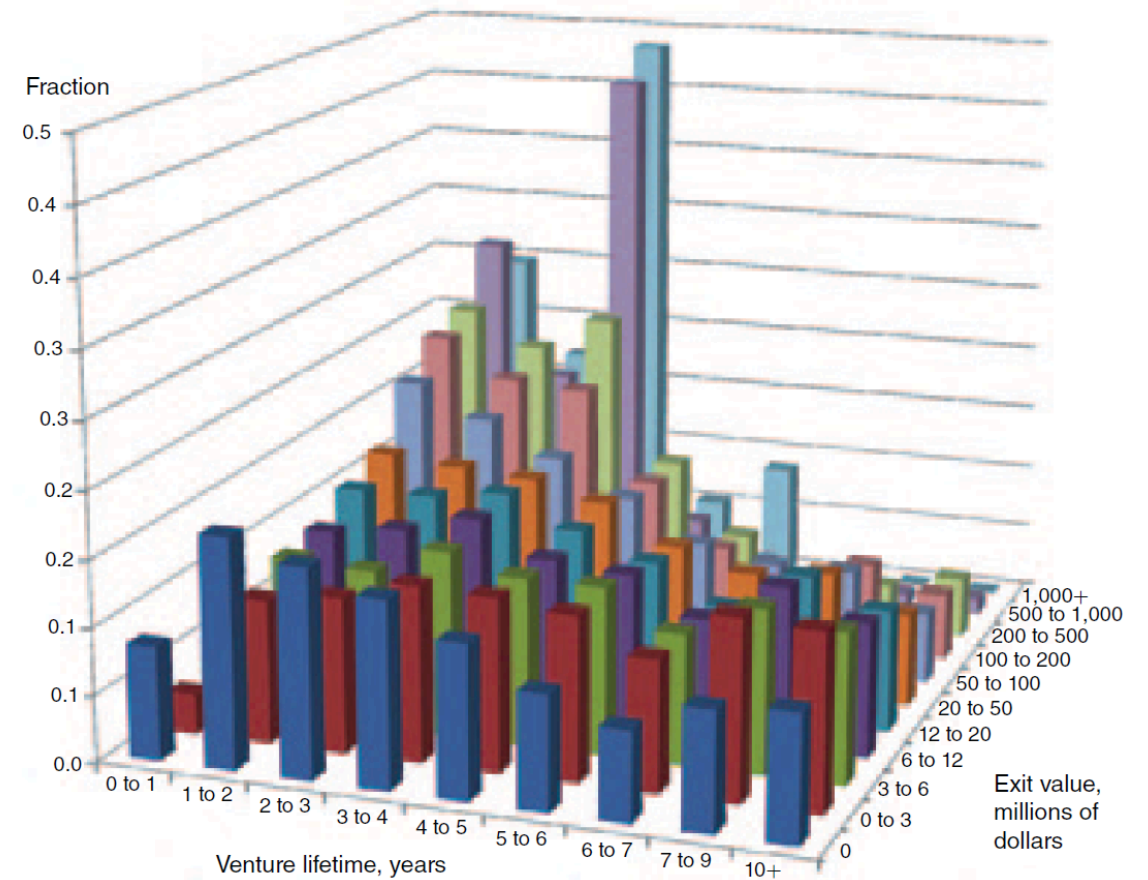


FIGURE 3

Source: Hall, R. E. & Woodward, S. E. (2010). The Burden of the Nondiversifiable Risk of Entrepreneurship. *American Economic Review*, 100(3), 1163–1194.

THE ENTREPRENEURSHIP PUZZLE

TABLE 3—CERTAINTY EQUIVALENT VALUE OF THE VENTURE OPPORTUNITY

Coefficient of relative risk aversion, γ	Pretax compensation at nonentrepreneurial job, thousands of dollars per year	Certainty equivalent of entrepreneurial opportunity, millions of dollars			
		Assets at beginning, millions of dollars			
		0.1	1	5	20
0	300	5.8	5.8	5.8	5.8
0	600	5.1	5.1	5.1	5.1
0	2,000	2.3	2.3	2.3	2.3
0.9	300	0.6	0.9	1.2	1.8
0.9	600	−0.2	0.3	0.7	1.2
0.9	2,000	−5.8	−4.0	−2.1	−1.6
2	300	−0.1	0.2	0.5	1.1
2	600	−1.8	−0.6	−0.1	0.5
2	2,000	−13.9	−8.9	−3.7	−2.2

Source: Hall, R. E. & Woodward, S. E. (2010). *The Burden of the Nondiversifiable Risk of Entrepreneurship*. *American Economic Review*, 100(3), 1163–1194.

THE QUESTIONS THESE FINDINGS RAISE

- Clearly, it is hard to explain why many people become entrepreneurs based on **expected financial returns alone**.
- **Who** becomes an entrepreneur, and **why**?
- What are the influences of particular **personal characteristics** and **environmental factors**?

POSSIBLE EXPLANATIONS

PREFERENCES / TRAITS-BASED

WHO IS WILLING TO ENTER

- Entrepreneurs “pay” for certain amenities (compensating differentials)
- Entrepreneurs like risk (and/or skewness)
- Entrepreneurs are too optimistic about the future
- Entrepreneurs are too confident about themselves or their projects

SKILLS-BASED

WHO ACTUALLY PERFORMS BETTER

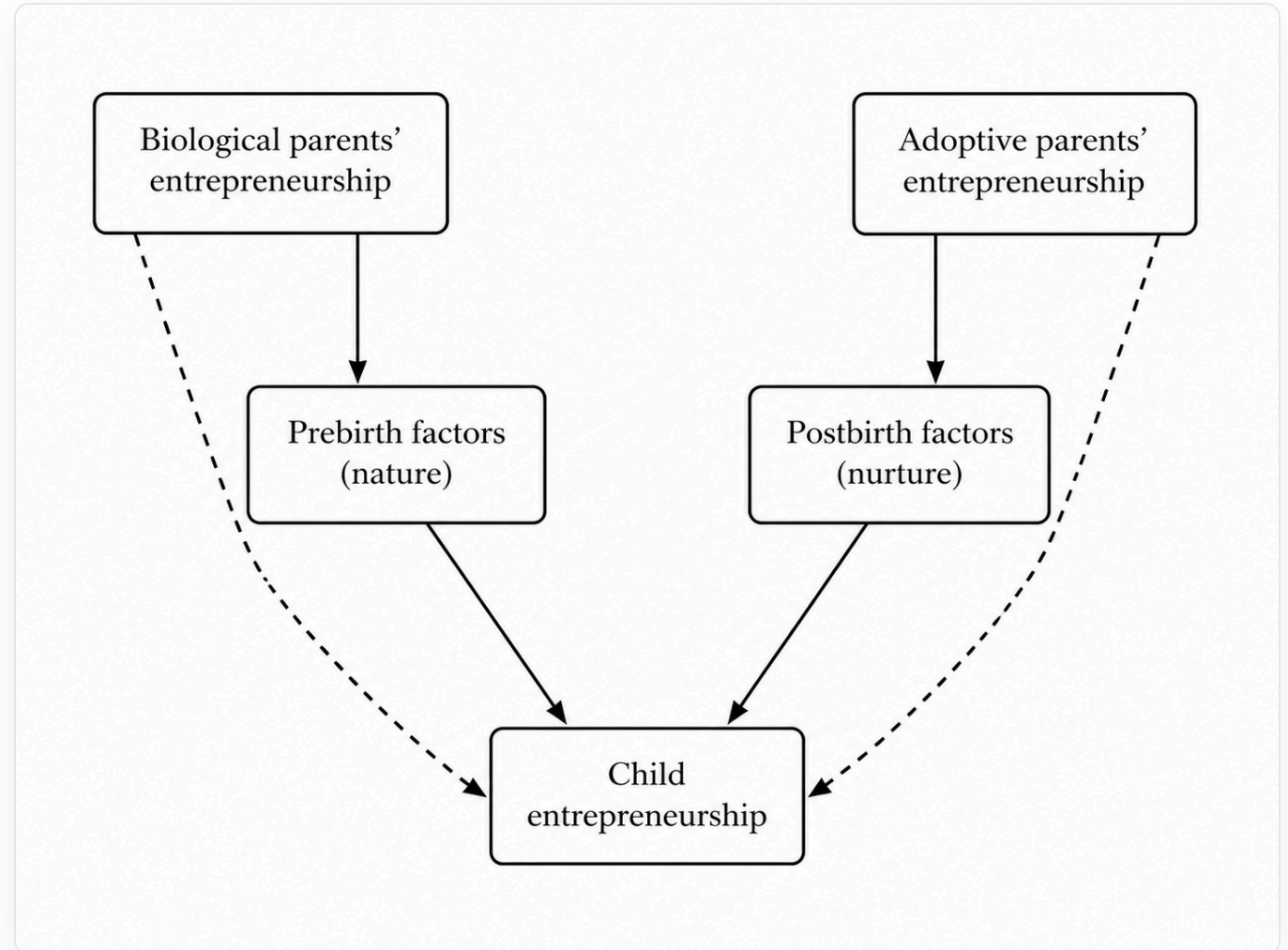
- Entrepreneurship rewards ability differently than wage work
- Entrepreneurs have a high portion of “entrepreneurial” human capital
- Entrepreneurs have a valuable social network

Before we examine these questions, we
first need to know...

ARE ENTREPRENEURS BORN OR MADE?

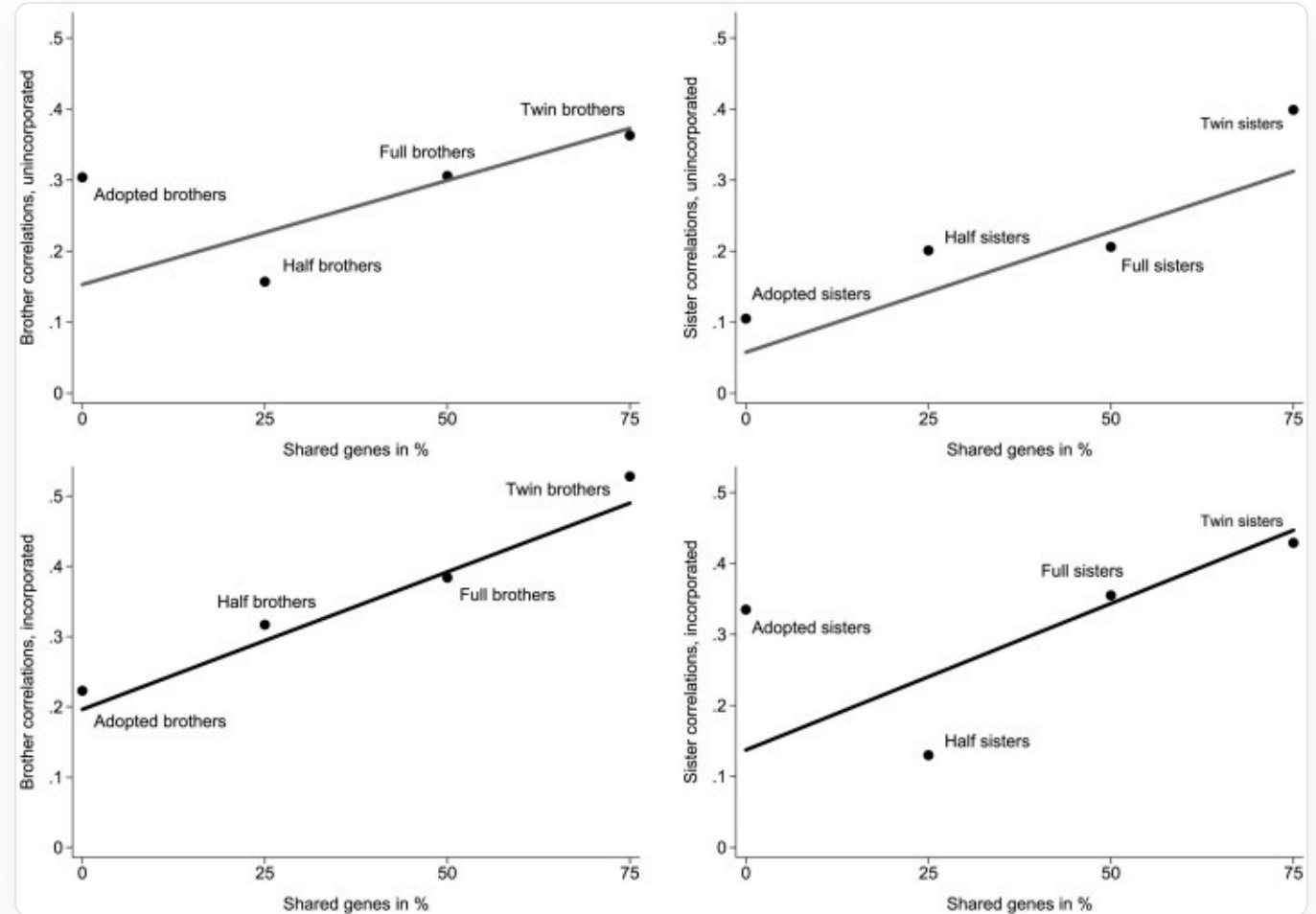
NATURE VS. NURTURE

- We have long known that having an **entrepreneurial parent** raises your odds of becoming an entrepreneur yourself.
- Is this **selection** (genes) or **treatment** (parental interactions)?
- By comparing **adopted vs. non-adopted** children, we can tease apart these influences.



NATURE VS. NURTURE

- **Lindquist et al. (2015)**: having entrepreneurial (biological) parents raises the likelihood of becoming an entrepreneur by **45–65%**. Two-thirds of this effect is **nurture**, one-third **nature**.
- **Vladasel et al. (2021)**: family influence matters more for **growth-oriented (incorporated)** than unincorporated entrepreneurs, with genes accounting for roughly **half** of this effect.



Lindquist, M. J., Sol, J., & Van Praag, M. (2015). Why do entrepreneurial parents have entrepreneurial children? *Journal of Labor Economics*, 33(2), 269–296.

Vladasel, T., Lindquist, M. J., Sol, J., & Van Praag, M. (2021). On the origins of entrepreneurship: Evidence from sibling correlations. *Journal of Business Venturing*, 36(5), 106017.

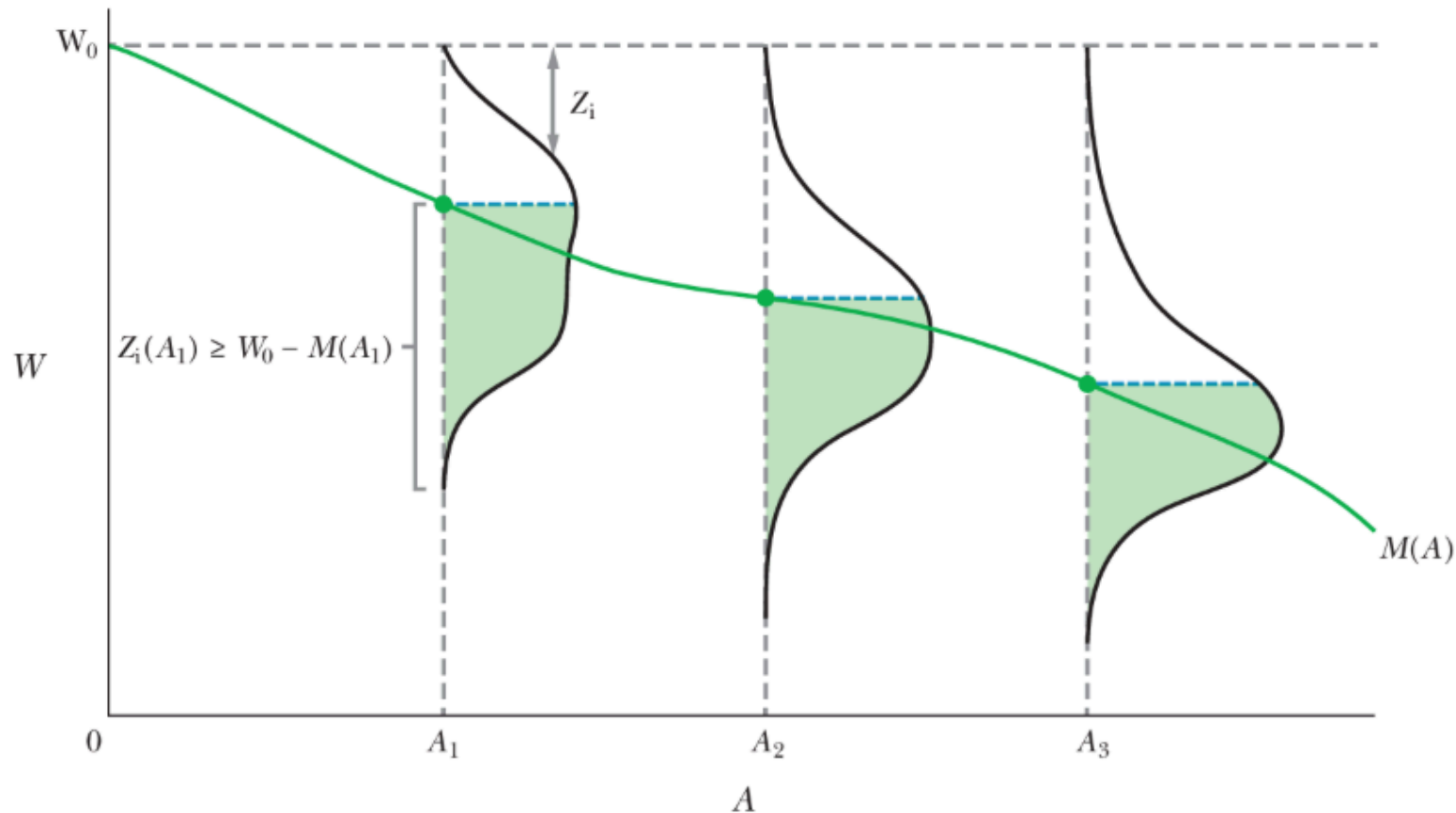
PREFERENCES

EXPLANATION 1: PAYING FOR WHAT THE JOB GIVES YOU

- **Compensating differentials:** if entrepreneurs earn *less* on average (Hamilton 2000 — ~35% lower median earnings after 10 years), they must be receiving **something else** in return.
- That “something else” is **non-pecuniary**: autonomy, flexibility, being your own boss, building something of your own, doing work you find meaningful.
- The financial “penalty” is then not a puzzle — it is the **price of a job you like**.

Hamilton, B. H. (2000). Does entrepreneurship pay? An empirical analysis of the returns to self-employment. Journal of Political Economy, 108(3), 604–631.

COMPENSATING DIFFERENTIALS (ROSEN, 1974; 1986)



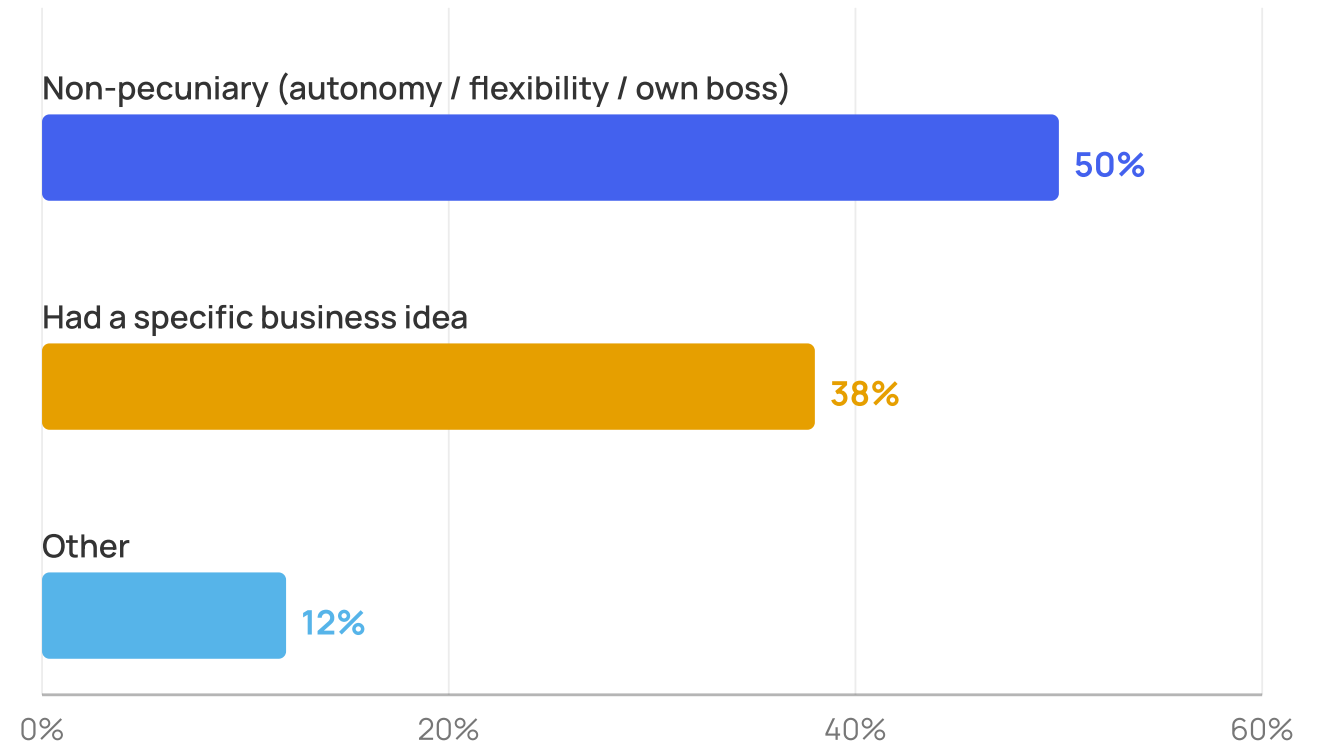
Source: Figure created by author.

Note: The horizontal axis shows levels of amenities A . The vertical axis shows wages W . W_0 is the wage for a job with zero amenities. For each discrete level of amenities A_1 , A_2 , and A_3 , the Z_i curve shows the distribution of preferences for the given amenity level across individuals. The $M(A)$ function shows the downward-sloping relationship between wages and amenities in equilibrium. All the workers in the shaded region, for whom $Z_i(A) \geq W_0 - M(A)$, value amenities by more than the market reduction in wages required to obtain them.

Source: Lavetti, Kurt. 2023. "Compensating Wage Differentials in Labor Markets: Empirical Challenges and Applications." *Journal of Economic Perspectives*, 37(3): 189–212.

WHY DO PEOPLE ACTUALLY START BUSINESSES?

- **Hurst & Pugsley (2011)**: most founders are **not** would-be high-growth innovators. Asked *why* they started, **non-pecuniary reasons dominate** (flexibility, being one's own boss).
- Most small-business owners **do not want to grow** and do not plan to innovate – plumbers, dentists, consultants, shopkeepers.
- “Having a good business idea” is cited by only about a **third** of founders.



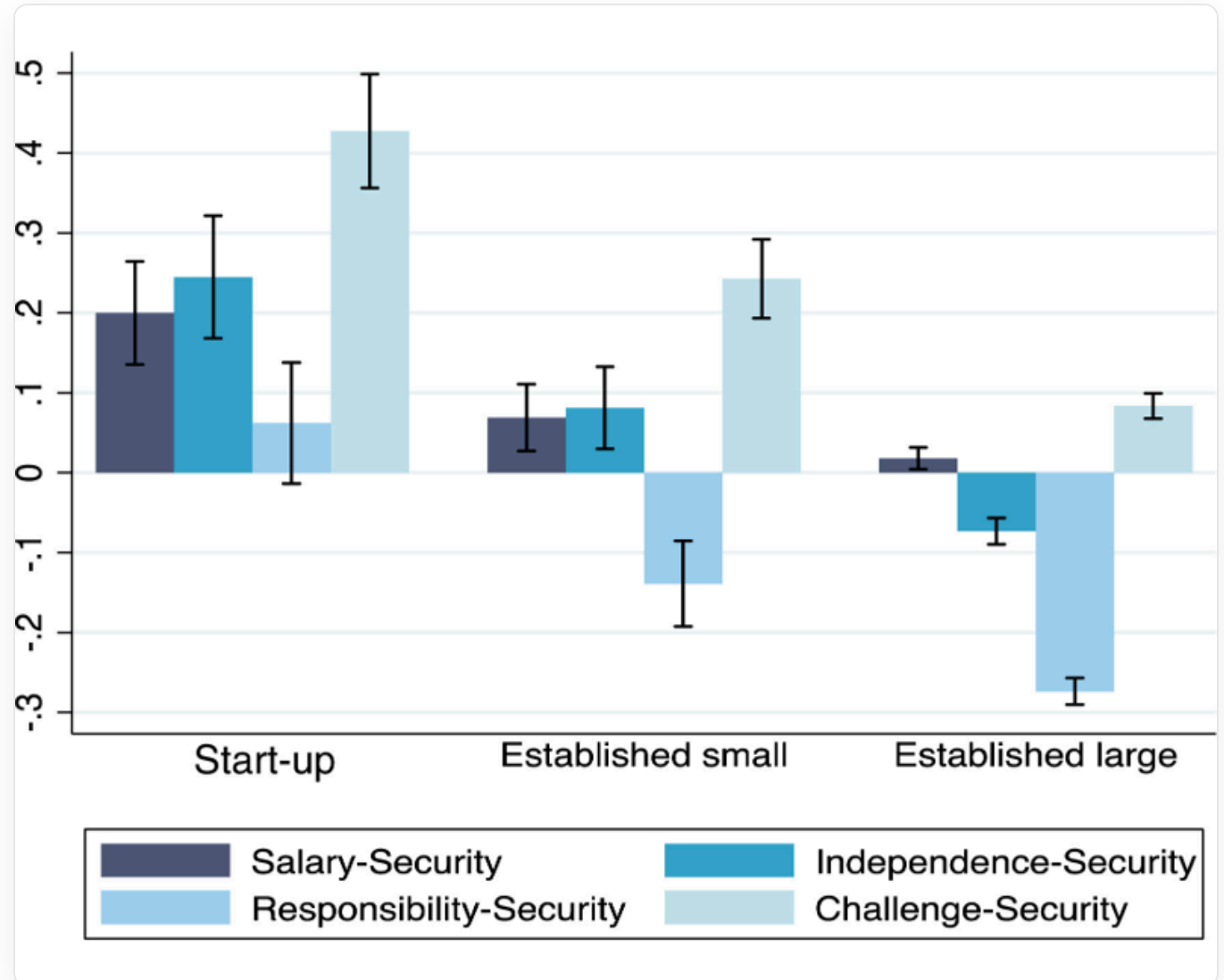
AUTONOMY IS WORTH PAYING FOR

- **Benz & Frey (2008)**: the self-employed report **higher job satisfaction** than employees – even after holding income and hours constant.
- This is **procedural utility**: people value not just *what* they get, but *how* the outcome comes about – being self-determined rather than told what to do.
- A concrete estimate of the compensating differential: people would **give up meaningful income** to keep the autonomy of working for themselves.

Benz, M., & Frey, B. S. (2008). Being independent is a great thing: Subjective evaluations of self-employment and hierarchy. Economica, 75(298), 362–383.

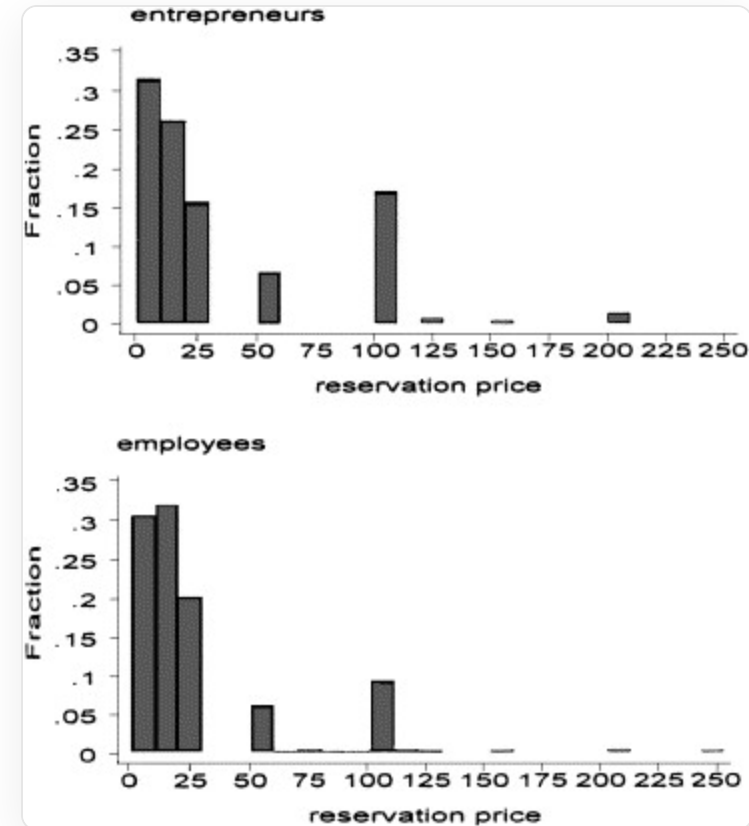
NOT ONLY THE MOMS AND POPS

- Tech-startup **joiners** value **independence and challenge** more than employees of established small or large firms.
- This “**fire in the belly**” is linked to a **higher rate of innovation** by startup employees.



EXPLANATION 2: RISK TOLERANCE

- Entrepreneurs are commonly portrayed as **risk takers** – so they may be “okay” with lower-than-expected returns, explaining part of the puzzle.
- What does the evidence say?
- **Cramer et al. (2002)** measured risk aversion via willingness to pay for a lottery ticket (10 tickets, single prize of 1,000 guilders). Conditional on wealth, this reservation price reflects risk aversion.
- Entrepreneurs are indeed **less risk averse** – but risk aversion is also highly correlated with (ex-post) wealth. Do risk-tolerant people become wealthier, or do the wealthy become risk-tolerant?



RISK TOLERANCE

- **Hvide & Panos (2014)** use observed financial behaviour (owning individual stocks) as a **revealed** measure of risk tolerance for ~400,000 Norwegians. Stockholders are **~50% more likely** to later start a firm.
- But more risk tolerance $\neq$ better firms: risk-tolerant entrepreneurs start firms with **~25% lower sales** and **~15% lower return on assets** than firms started by the more risk-averse.
- So risk tolerance predicts **who enters** — but if anything it predicts **worse** outcomes, not better ones.

Hvide, H. K., & Panos, G. A. (2014). Risk tolerance and entrepreneurship. Journal of Financial Economics, 111(1), 200–223.

RISK TOLERANCE

Cramer et al. (2002) and Hvide & Panos (2014) pool *all* entrepreneurs – but what about the more **growth-oriented, innovative** types?

3.60

SCIENCE & TECH STARTUPS

Bailey et al. (2026), scale 1 (most risk averse) – 4 (most risk loving)

1.72

GENERAL POPULATION

Barsky et al. (1997) population mean, same scale

How does this compare to your risk preference score?

Bailey-Rihawi, E., Fehder, D. C., Floyd, E., Hochberg, Y. V., & Lee, D. (2026). Learning to Quit? A Multi-Year, Multi-Site Field Experiment with Innovation-Driven Entrepreneurs.

RISK TOLERANCE

- **Levine & Rubinstein (2017)** use a different proxy for risk tolerance: “**illicit**” **behaviour as teenagers** (damaging property, fighting, shoplifting, robbery, using force, assault or threats, drug use, dealing, gambling, being stopped by police, being charged).
- **Incorporated** entrepreneurs (their proxy for growth orientation) are **twice as likely** as salaried workers to report having taken something by force as youths, **~40% more likely** to have been stopped by police, and score **0.2 SD higher** on an overall illicit-activity index.
- The unincorporated also engage in more illicit activity than salaried workers — but the **incorporated engage in still more**.

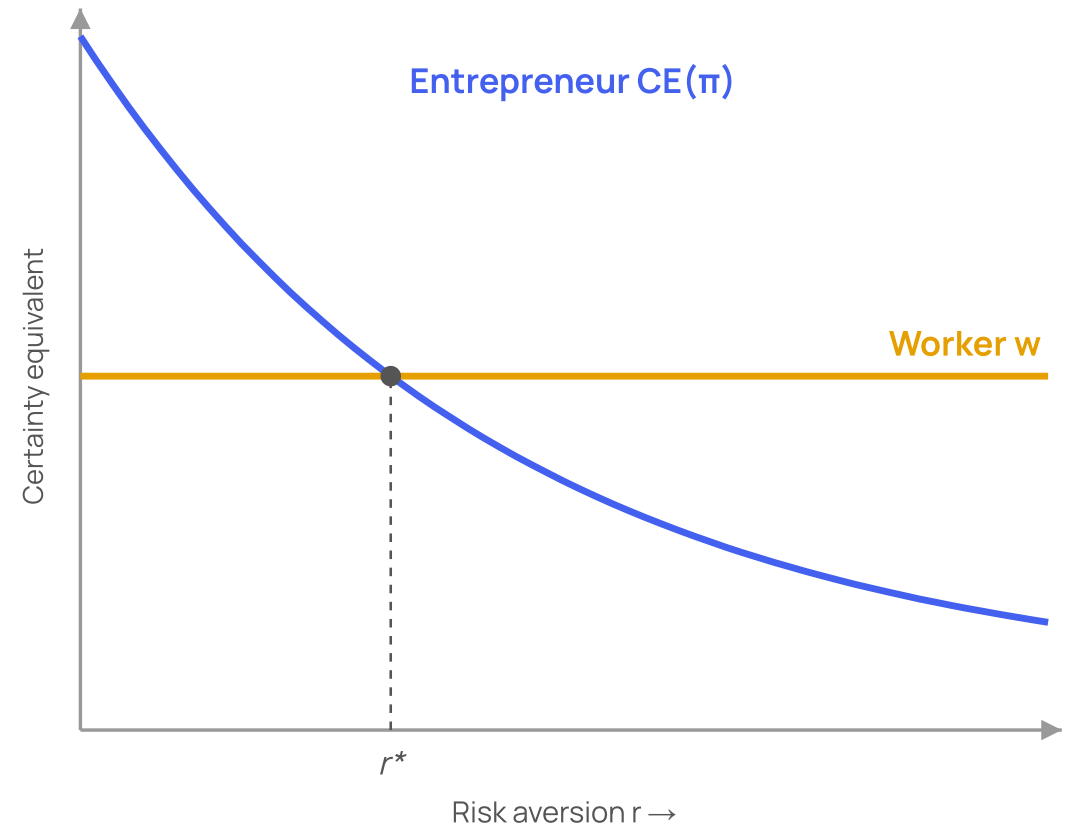
Levine, R., & Rubinstein, Y. (2017). Smart and Illicit: Who Becomes an Entrepreneur and Do They Earn More? Quarterly Journal of Economics, 132(2), 963–1018.

RISK TOLERANCE — KIHLSSTROM & LAFFONT (1979)

- Suppose people differ only in risk aversion r . Each chooses to be a worker or an entrepreneur.
- **Worker:** earns the certain wage $w \rightarrow$ utility $u(w)$.
- **Entrepreneur:** hires labour L and runs a risky technology $x = \theta f(L)$ (f concave, shock θ , $E[\theta]=1$). Profit $\pi = \theta f(L) - wL$ is uncertain.
- You enter iff the risky venture beats the wage:

$$E[u(\pi)] \geq u(w)$$

- The risk premium grows with r , so there is a single cutoff r^* : people with $r < r^*$ become entrepreneurs, those with $r > r^*$ become workers.



RISK TOLERANCE — KIHLSSTROM & LAFFONT (1979)

FITS THE EVIDENCE

WHERE K-L DOES WELL

- Entrepreneurs are, in general, **more risk-tolerant**.
- Innovation-driven / growth-oriented types are more risk-tolerant than subsistence types.

HARD TO EXPLAIN

WHERE K-L STRUGGLES

- More risk-tolerant **underperform** the more risk-averse entrepreneurs (Hvide & Panos, 2014).
- No **ability**, endogenous risk aversion (Camerer et al., 2002), or innovation — entrepreneurs are for **skills** purely profit-driven.

RISK TOLERANCE — KIHLSSTROM & LAFFONT (1979)

- An economy-wide rise in risk aversion (a pandemic, an oil crisis, ...) lowers **both** the number of entrepreneurs **and** wages.
- K-L assume entrepreneurs bear **all** the costs of failure. What if there is **limited liability**? Payoff becomes $\tilde{\pi} = \max(\theta f(L) - wL, 0)$ — losses beyond zero fall on others.
 - More risk-averse people now enter entrepreneurship (though this alone doesn't say what they build).
 - Conditional on entering, they run **larger firms** and hire more: capping the downside removes the worst-case states that made scale costly, so the same r supports a bigger L .
 - Wages **rise** (more entrants = more labour demand), partly — but not fully — offsetting the scale effect.

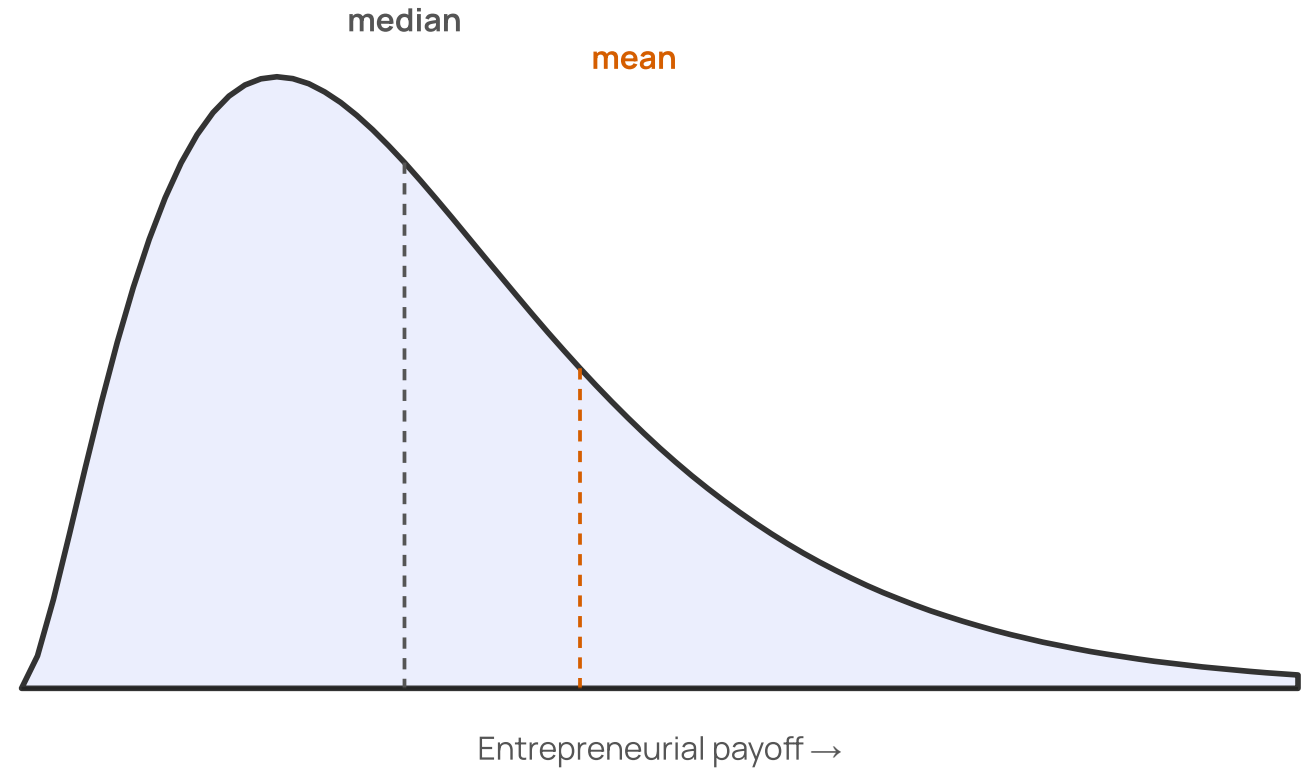
THE FINE LINE OF LIMITED LIABILITY

- **Without** limited liability, few people become entrepreneurs – and those who do may avoid “aiming big” to limit the risk of catastrophic personal losses.
- **With** limited liability, however, entrepreneurs may take **excessive risks** if they believe they cannot be held personally accountable.



A TASTE FOR SKEWNESS: ENTREPRENEURSHIP AS A LOTTERY

- Startup payoffs are **right-skewed**: most ventures fail or earn little; a few earn spectacular returns.
- People routinely **pay for positive skewness** — buying lottery tickets and betting long shots even at negative expected value (the “favorite-long-shot bias”; Golec & Tamarkin 1998).
- If founders **value the small chance of a huge win**, they will enter even when the *median* and *mean* returns are unattractive.



EXPLANATION 3: ENTREPRENEURS ARE OPTIMISTS

WHAT OPTIMISM IS

- **Optimism** = the consistent belief that you are *less* likely than others to suffer bad events and *more* likely to experience good ones (Weinstein, 1980). “Good things will happen to *you*.”
- It is strongest when (a) the event feels **controllable** and (b) there is **commitment** or emotional investment in the outcome.
- **Cooper, Woo & Dunkelberg (1988)** surveyed ~3,000 new founders:
 - **81%** put their *own* chances at 7-in-10 or better; **33%** said a **perfect 10-in-10**.
 - Actual survival is far lower — and estimates for “*a business like yours*” dropped sharply.

THE INNOVATION-DRIVEN TYPES

- **Bailey et al. (2026)**: a max optimism score of almost **20 (out of 24)**, vs **14.33** for the standard lab population (Scheier, Carver & Bridges, 1994).
- **Åstebro et al. (2007)**: independent inventors are significantly more optimistic than the general population.
 - **1/3** keep spending money and **1/2** keep spending time on projects *after* receiving highly diagnostic advice to cease.
 - The highly optimistic keep spending even after explicitly **negative** advice.

Cooper, A. C., Woo, C. Y., & Dunkelberg, W. C. (1988). Entrepreneurs' perceived chances for success. *Journal of Business Venturing*, 3(2), 97–108.

Åstebro, T., Jeffrey, S. A., & Adomdza, G. K. (2007). Inventor perseverance after being told to quit: the role of cognitive biases. *Journal of Behavioral Decision Making*, 20, 253–272.

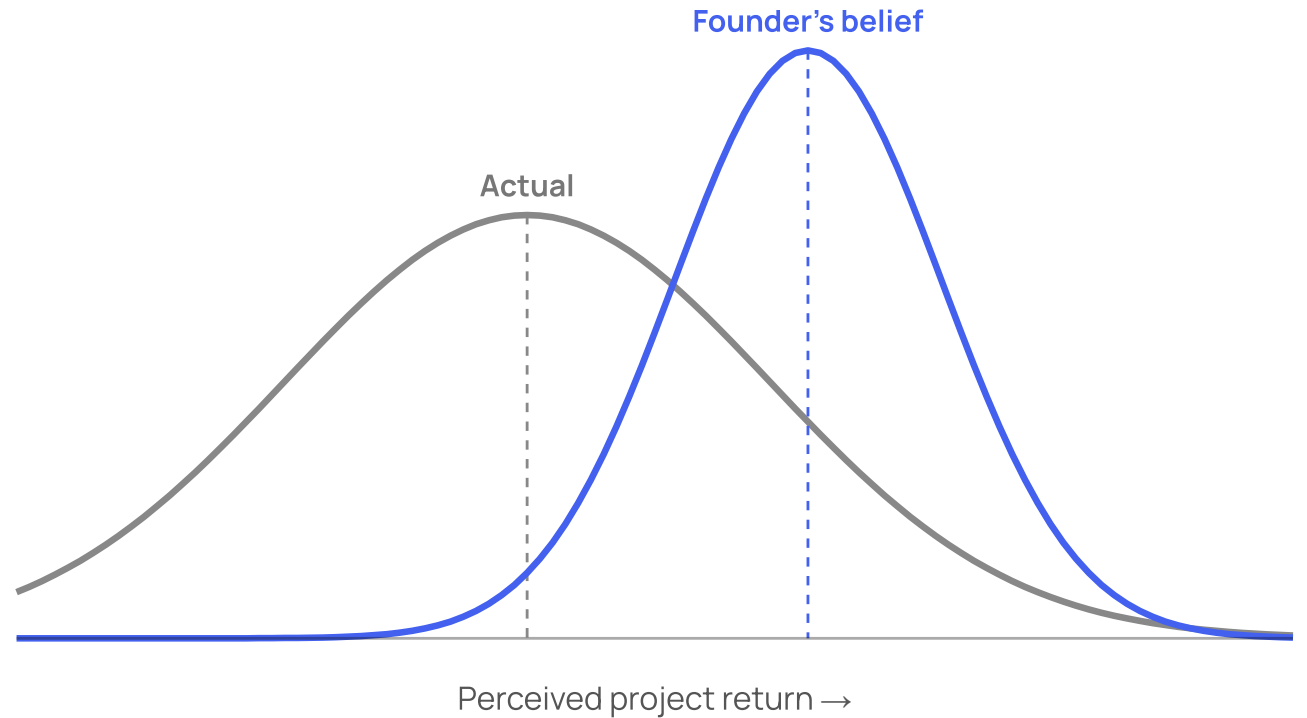
OPTIMISM HAS CONSEQUENCES

- **Landier & Thesmar (2009)**: French founders make **systematic upward errors** — they overestimate future sales and hiring.
- More educated founders, and those with their **own** idea, are the most optimistic.
 - Those with better non-entrepreneurial outside options only enter when they are *really* optimistic (self-selection).
- Optimists **self-select into short-term debt**: they expect to refinance on better terms soon, creating a separating equilibrium between optimists and realists.
- Mild optimism may even help persistence — but it means expected returns are systematically *overstated* at entry.

Landier, A., & Thesmar, D. (2009). Financial contracting with optimistic entrepreneurs. The Review of Financial Studies, 22(1), 117–150.

EXPLANATION 4: OVERCONFIDENCE — THREE FLAVOURS

- People enter because they **subjectively perceive the return distribution too favorably** when evaluating their *own* project.
- **Moore & Healy (2008)** separate three biases that often get lumped together:
 - **Over-estimation** — thinking you (or your project) are better than you really are.
 - **Over-placement** — thinking you are better than *others* (“most startups fail, but not mine”).
 - **Over-precision** — being too certain your forecasts are right.



OVERCONFIDENCE AND EXCESS ENTRY

THE CORE EXPERIMENT

- **Camerer & Lovallo (1999)**: when payoffs depend on **skill**, people **over-enter** markets; when payoffs are random, entry is rational (over-placement).
 - Subjects entered a “market” where payoffs depended on their *rank*; in the skill condition, rank came from a trivia quiz.
- **“Reference-group neglect”**: entrants focus on their own skill and forget they compete against other confident entrants.

FOUNDERS IN THE FIELD

- **Busenitz & Barney (1997)**: founders are more overconfident than managers in large firms – equally likely to be wrong, far more sure they are right.
- **Koellinger, Minniti & Schade (2007)**: across 18 countries, subjective (often biased) confidence is one of the strongest correlates of starting a business.
- Not all overconfidence causes excess entry: **Herz et al. (2014)** find **over-precision** leads to *less* experimentation (tunnel vision).

**OPTIMISM AND
OVERCONFIDENCE:
A BUG — OR A
FEATURE?**

DO YOU HAVE AN “ENTREPRENEURIAL PERSONALITY”?

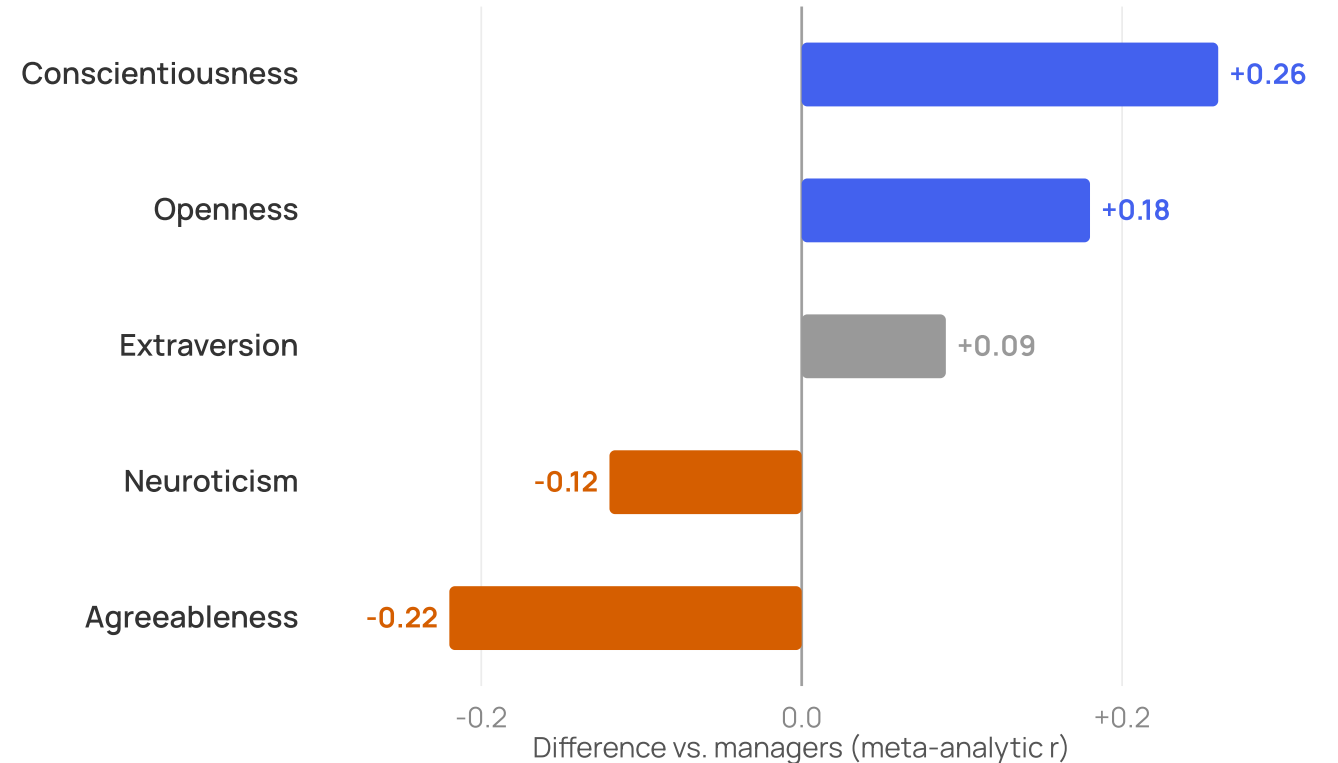


Take the test and find out – scan the code or follow the link:

jmahieu.github.io/entrepreneurial-personality-traits

WHAT THE BIG FIVE SAY

- **Zhao & Seibert (2006)**, meta-analysis of entrepreneurs vs. managers. Entrepreneurs score:
- **Higher** on Conscientiousness and Openness to experience.
- **Lower** on Agreeableness and Neuroticism (i.e. more emotionally stable, more competitive).
- **No reliable difference** on Extraversion.



Zhao, H., & Seibert, S. E. (2006). The Big Five personality dimensions and entrepreneurial status: A meta-analytical review. Journal of Applied Psychology, 91(2), 259–271. (Effects are small; magnitudes approximate.)

TRAITS PREDICT *ENTRY* BETTER THAN *SUCCESS*

- Zhao, Seibert & Lumpkin (2010), meta-analysis: the Big Five relate to entrepreneurial **intentions** (multivariate $R \approx .36$) somewhat more strongly than to entrepreneurial **performance** ($R \approx .31$).
- Some predictors of *wanting to* and *becoming* an entrepreneur (e.g. Agreeableness, risk propensity) **do not predict how well the venture does.**

Zhao, H., Seibert, S. E., & Lumpkin, G. T. (2010). The relationship of personality to entrepreneurial intentions and performance: A meta-analytic review. Journal of Management, 36(2), 381–404.

SO HOW MUCH DO TRAITS REALLY EXPLAIN?

- **Kerr, Kerr & Xu (2018)** review the modern evidence: traits like internal locus of control, risk-taking, tolerance of ambiguity and overconfidence **do matter** — but effects are **modest**, and the field is full of **disagreement**, because “entrepreneurship” is so heterogeneous.
- **Gartner (1988)**: maybe we are asking the wrong question. **“Who is an entrepreneur?” is the wrong question** — entrepreneurship is not a personality type, it is an **activity**: creating an organization. Ask *what entrepreneurs do*, not *who they are*.

Kerr, S. P., Kerr, W. R., & Xu, T. (2018). Personality traits of entrepreneurs: A review of recent literature. Foundations and Trends in Entrepreneurship, 14(3), 279–356. · Gartner, W. B. (1988). “Who is an entrepreneur?” is the wrong question. American Journal of Small Business, 12(4), 11–32.

**SKILLS, ABILITY &
HUMAN CAPITAL**

YES, I CAN (?)

- So far we explained the puzzle by saying entrepreneurs **like** the nature of the work.
- These explanations mostly take returns as **given / exogenous** – e.g. Kihlstrom & Laffont (1979), where entrepreneurial performance is *random*.
- But entrepreneurs can also **influence their performance** (why else the high internal locus of control?).
- What if traits and skills **impact performance**? Low *average* returns may hide high returns for a skilled minority – the puzzle is about **heterogeneity**, not a universally bad deal.

FOCUS ON TWO EXPLANATIONS

- Entrepreneurship rewards ability more steeply than wage work
- Entrepreneurs carry valuable, unique traits and skills

EXPLANATION 5: ENTREPRENEURSHIP

REWARDS ABILITY

- People differ in ability a and sort into the occupation where their talent pays best – **comparative advantage**. Potential income rises with ability in each:

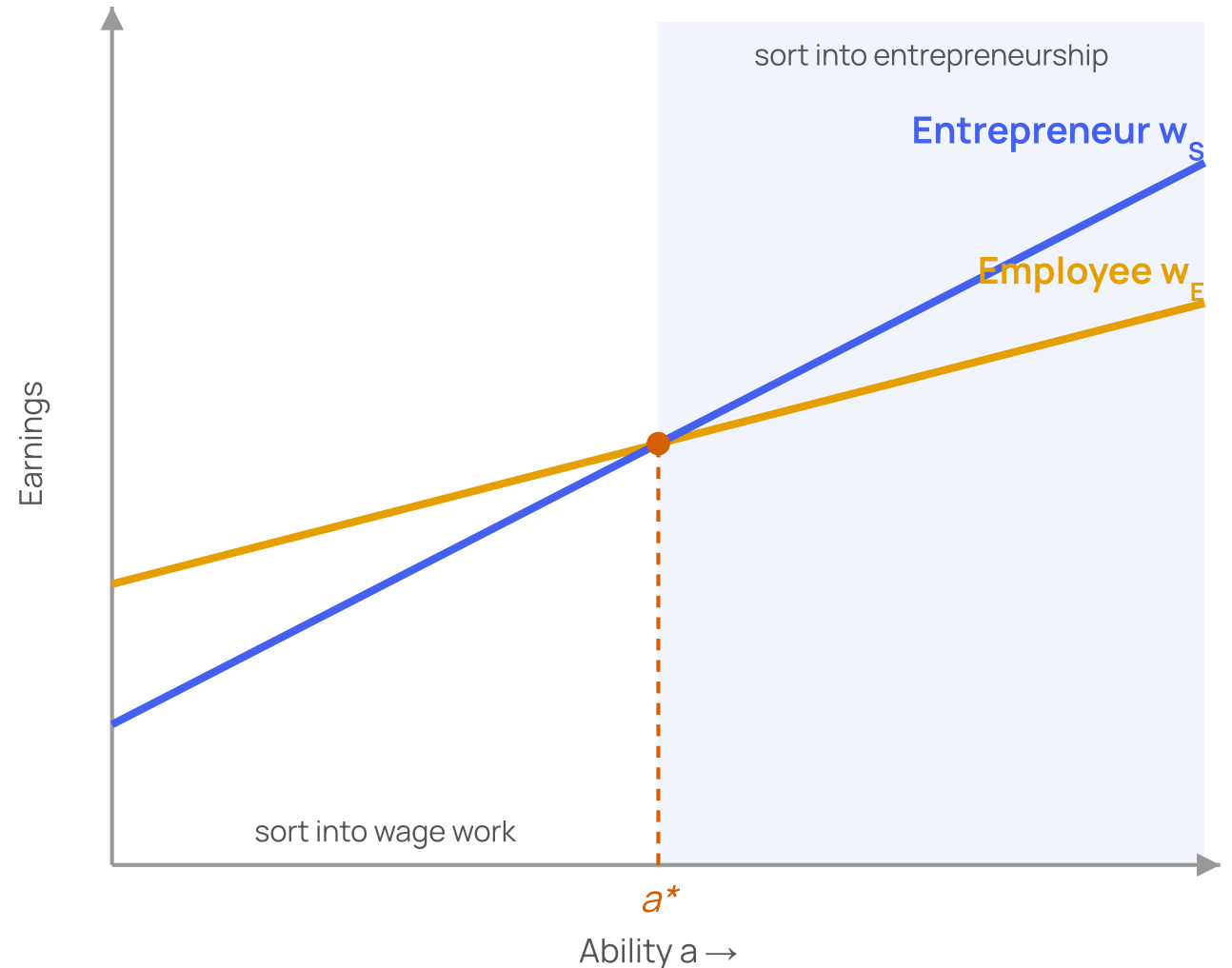
Employee: $w_E(a) = \mu_E + \gamma_E a$

Entrepreneur: $w_S(a) = \mu_S + \gamma_S a$

- **Choice rule:** become an entrepreneur iff $w_S(a) \geq w_E(a)$.
- Two assumptions drive the result:
 - $\gamma_S > \gamma_E$ – the entrepreneur is the **residual claimant**, keeping the marginal return to her talent (a wage compresses it). Ability is rewarded *more steeply*.
 - $\mu_S < \mu_E$ – but she gives up the safe baseline wage, so at **low ability** entrepreneurship pays *less*.

ENTREPRENEURSHIP REWARDS ABILITY

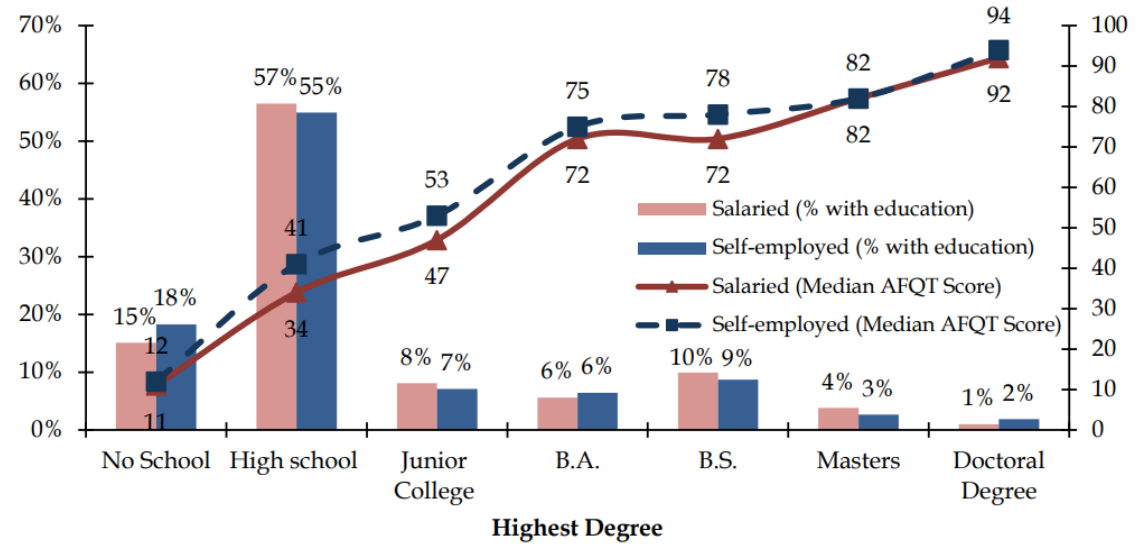
- The **gain** from entrepreneurship rises with ability:
 $\Delta(a) = (\mu_S - \mu_E) + (\gamma_S - \gamma_E)a$.
- Since $\gamma_S > \gamma_E$, there is a single **ability threshold**
 $a^* = (\mu_E - \mu_S) / (\gamma_S - \gamma_E)$: $a > a^* \rightarrow$ entrepreneur, $a < a^* \rightarrow$ employee $\Rightarrow$ **positive sorting on ability**.
- **Predictions:** the ablest self-select into entrepreneurship *and* earn more there; entrepreneurial earnings are **more dispersed**.



ENTREPRENEURSHIP REWARDS ABILITY

- **Levine & Rubinstein (2017)**: incorporated entrepreneurs have higher cognitive ability (AFQT) than employees *and* than unincorporated entrepreneurs.
- **Hegde & Tumlinson (2021)**: entrepreneurs have higher cognitive ability at **each level of education** (both incorporated and unincorporated).
- But they do **not** find higher *returns* to cognitive ability for entrepreneurs.

Figure 2: Ability scores and education, by occupational status



The figure compares median AFQT scores for self-employed and salaried individuals at various levels of individuals' educational qualifications. The vertical bars indicate the percentage of person-year observations during which the worker had attained the corresponding educational qualification by employment status. The figure is based on 9,922 person-year observations of self-employed individuals and 99,099 person-year observations of wage employees in the NLSY panel.

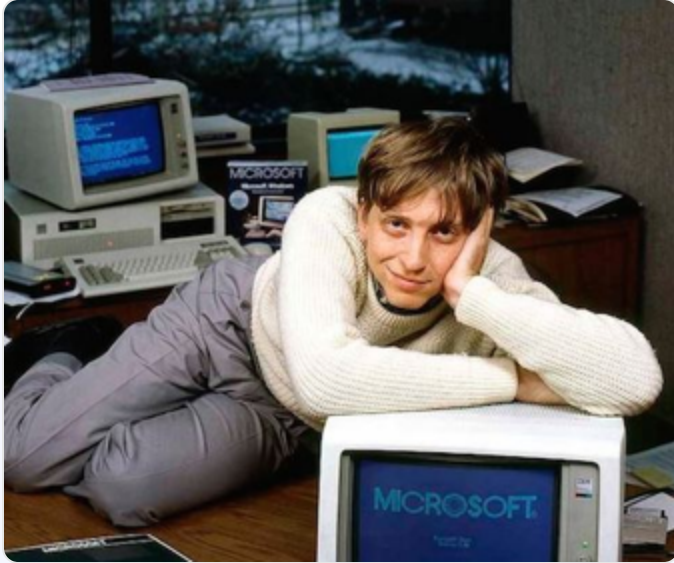
ENTREPRENEURIAL HUMAN CAPITAL

- These explanations treat ability as a **one-dimensional** “more is better.” But is human capital in entrepreneurship really that simple?
- Ability is also usually treated as **given** — something you are born with. What about the human capital that can be “**made**”?

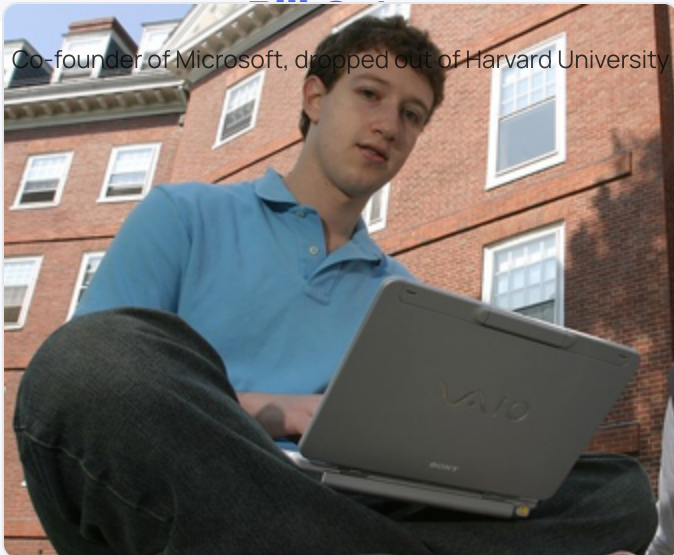
DOES EDUCATION PAY FOR ENTREPRENEURS?



DOES EDUCATION PAY FOR ENTREPRENEURS?



Steve Jobs



Co-founder of Microsoft, dropped out of Harvard University

Mark Zuckerberg

Co-founder of Facebook, dropped out of Harvard University



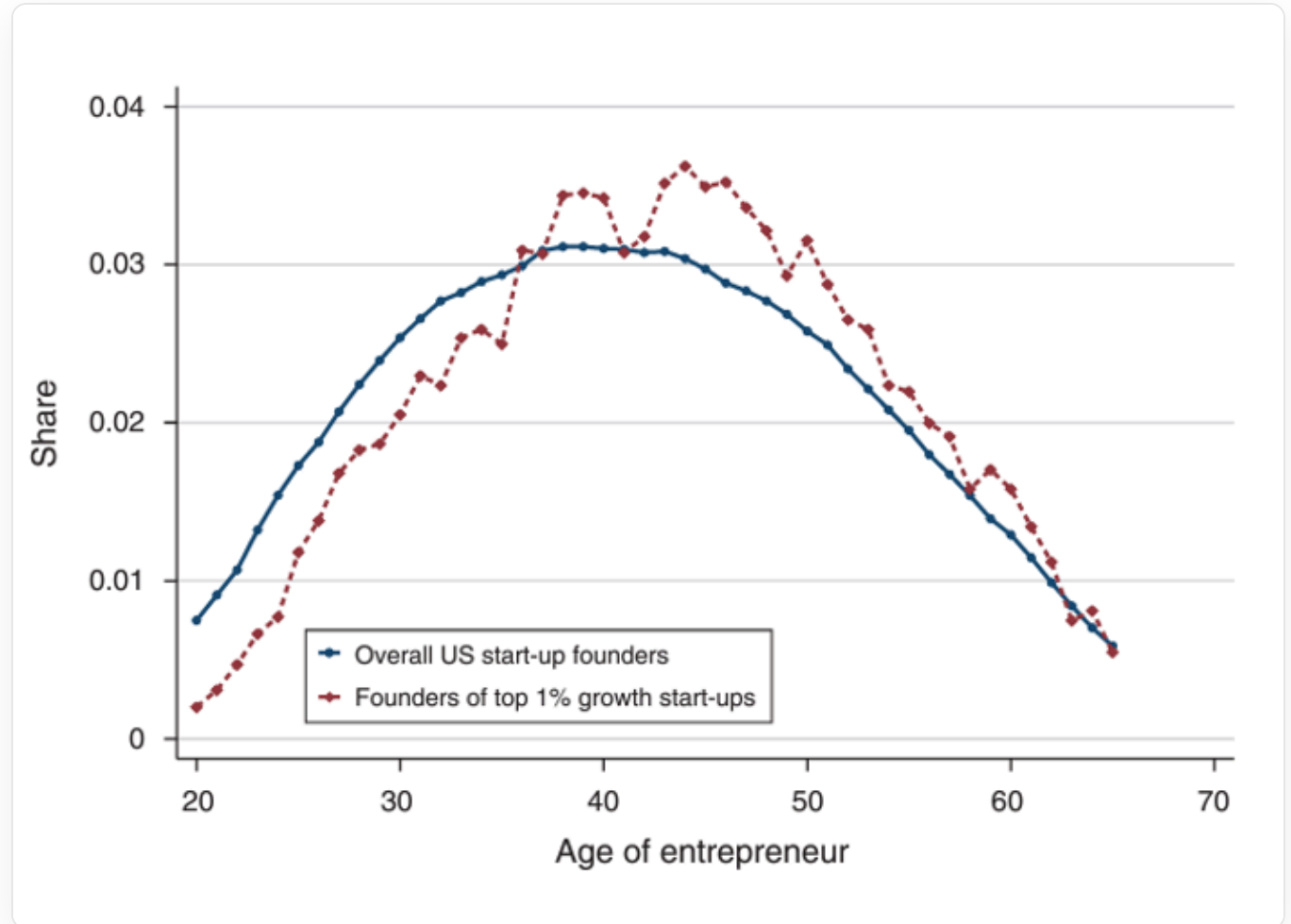
Co-founder of Apple, dropped out of Reed College

Elizabeth Holmes

Founder of Theranos, dropped out of Stanford University

ARE FOUNDERS YOUNG DROPOUTS?

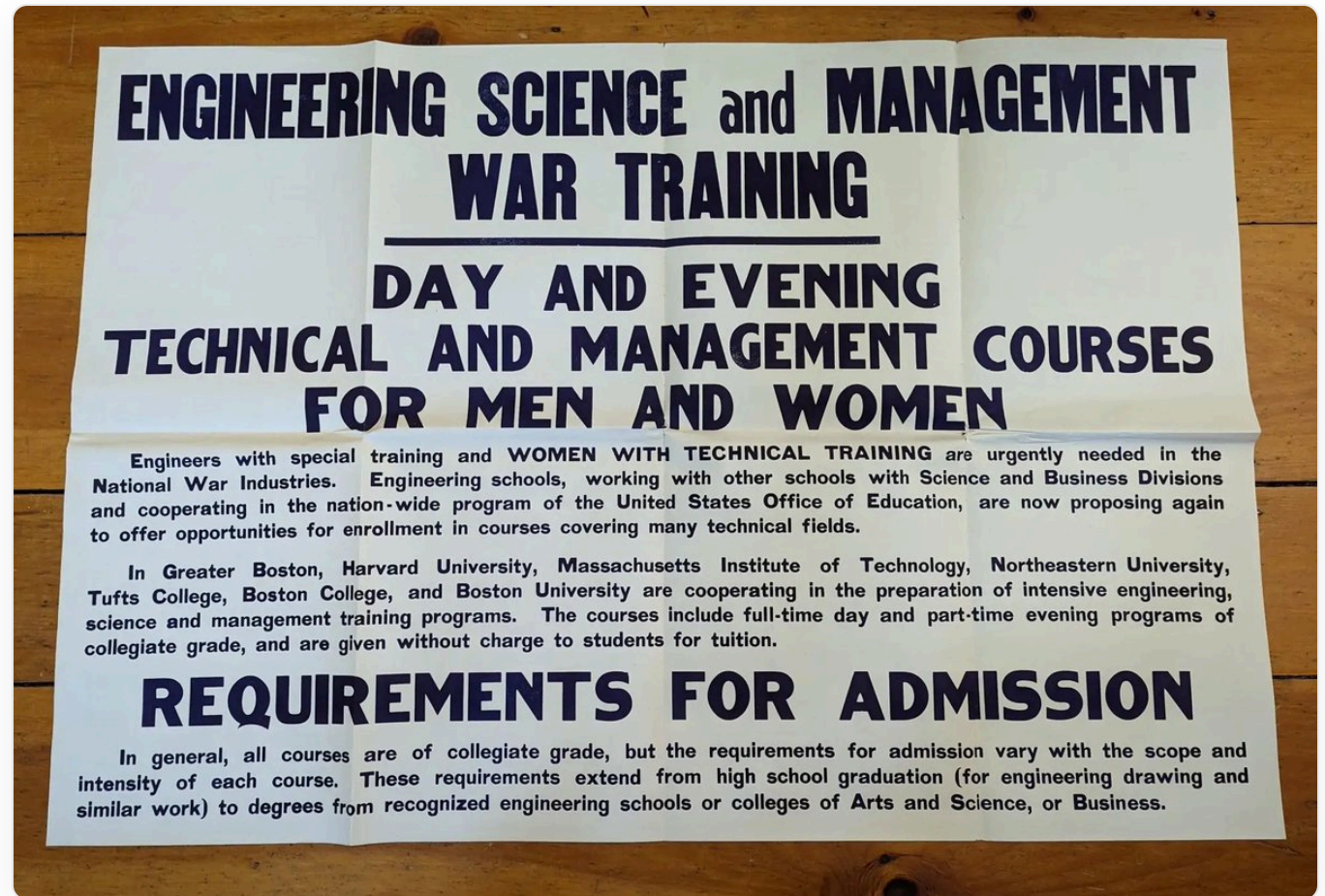
- The pervasive image: the **twenty-something** who **dropped out of college** to chase a startup dream and became an ultra-successful billionaire.
- Yet most highly successful entrepreneurs are **middle-aged (40–50)**, not young.
- Most startups are founded by individuals with **prior work experience** (see next lecture).



DOES EDUCATION PAY FOR ENTREPRENEURS?

- **Giorcelli (2024)** studies the “Engineering, Science and Management War Training” (ESMWT) program — MBA-style education for middle managers and production supervisors at U.S. war industrial facilities in WWII.
- ESMWT attendance **systematically increased** managers’ entrepreneurial activity. Effects were **strongest for nonwhite and female** participants — why?
- Not just a signaling effect (“badge of honor”) — participation led to:
 - **Better firm performance** (less absenteeism, late orders, injuries, scrap) via adoption of managerial practices
 - **Network building** with managers of other facilities, raising the likelihood of co-founding a business together

Giorcelli, M. (2024). The Effects of Business School Education on Manager Career Outcomes.

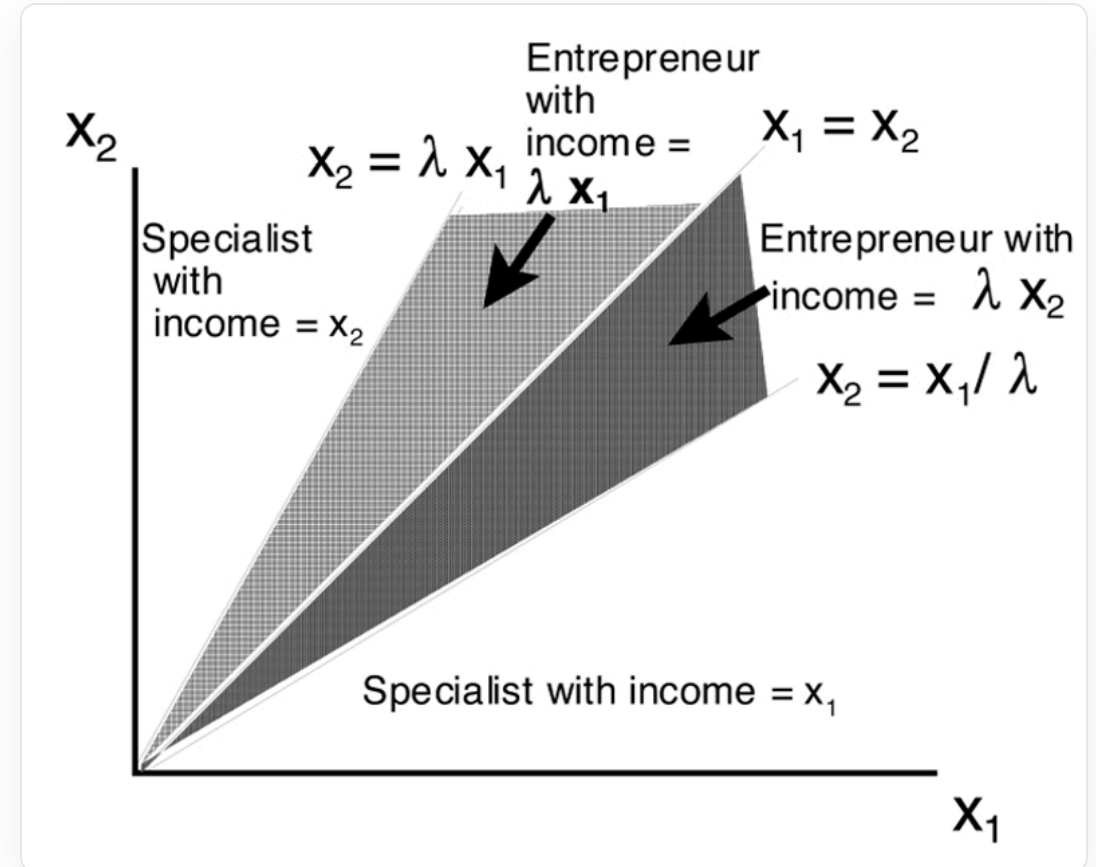


EXPLANATION 6: THE 'JACK-OF-ALL-TRADES'

- **Lazear (2004, 2005)**: running a firm needs a bit of everything – entrepreneurs are **generalists**, not specialists.
- **Balance beats depth**: your earnings as a founder are limited by your **weakest** essential skill.

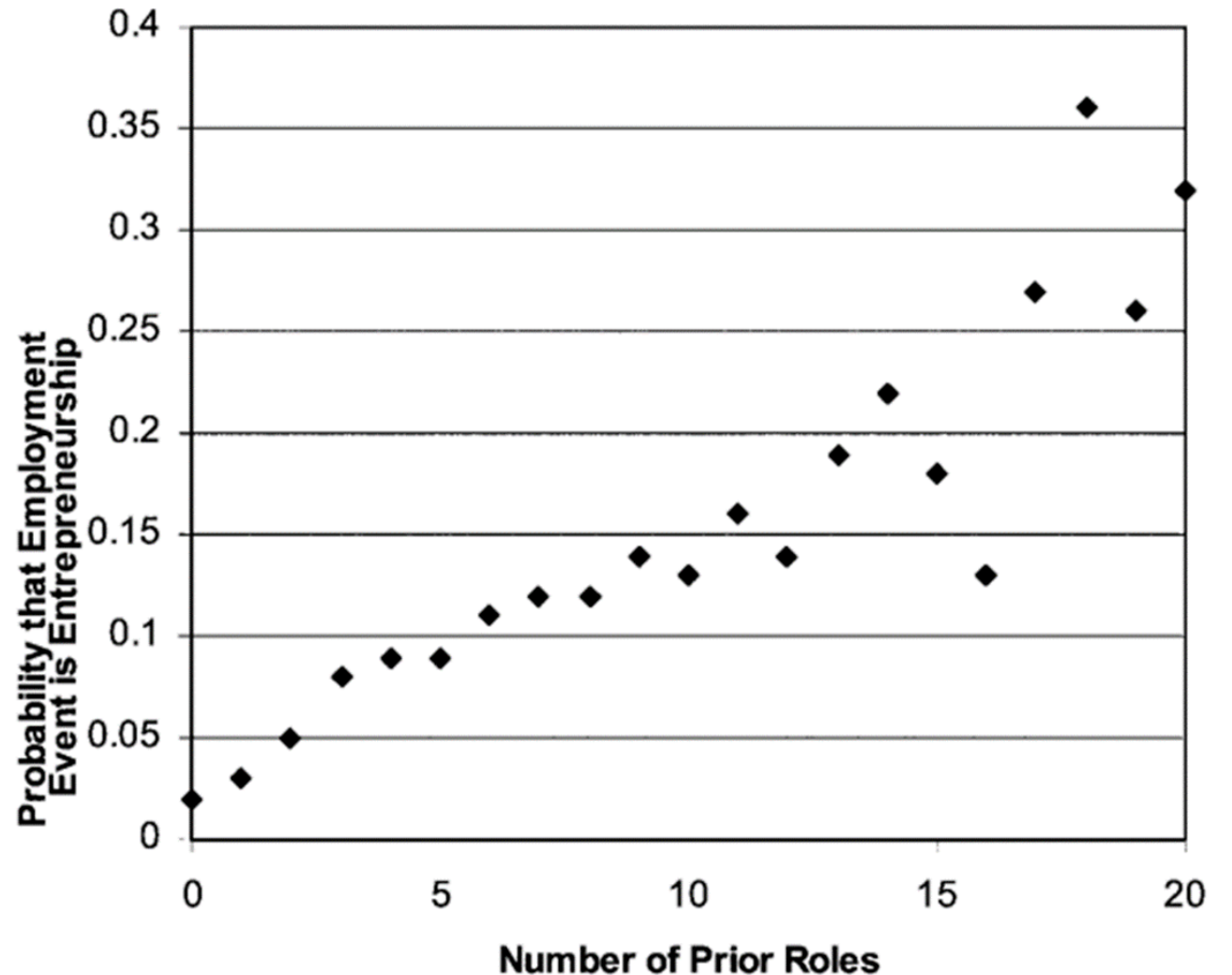
EMPLOYEE $w = \max\{x_1, x_2\}$
specializes – paid for their *strongest* skill

ENTREPRENEUR $\pi = \lambda \min\{x_1, x_2\}$
needs both – only as good as the *weakest*



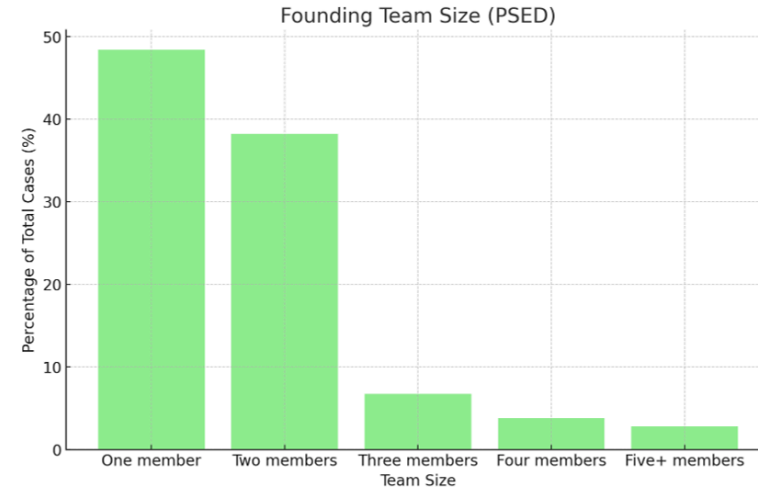
EXPLANATION 6: THE 'JACK-OF-ALL-TRADES'

- People with more **diverse education, prior work roles or fields** are more likely to become (better) entrepreneurs.
- Top management teams with more **diverse experience** have more success in accessing venture capital.

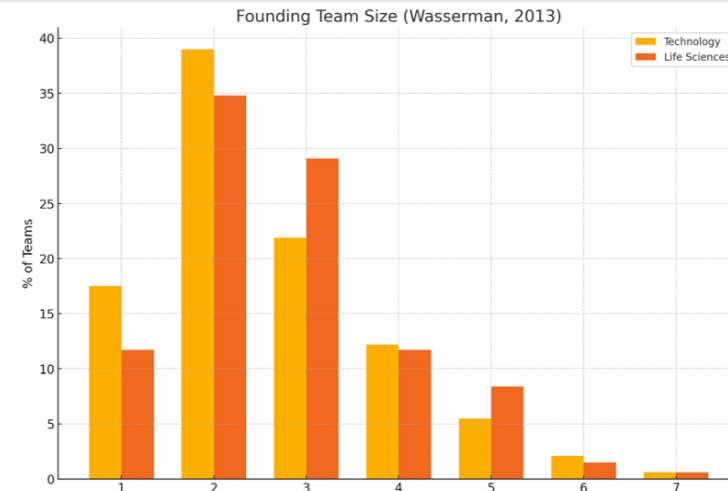


INNOVATION-DRIVEN ENTREPRENEURSHIP IS A *TEAM* SPORT

- Traditional small business is mostly **solo**: the large majority have **no employees** and a single owner.
- Innovation-driven ventures are typically **team-founded** — no one person holds all the specialized skills a differentiated product needs.
- The founding **team** is how ventures satisfy Lazear's balance requirement *without* every individual being a generalist.



Based on data from Ruef, Aldrich & Carter (2003), *American Sociological Review*, 68(2), 195–222.



Based on data from Wasserman (2013), *The Founder's Dilemmas*, Ch. 3.

RECONCILING THE PUZZLE

- Low **average** returns are fully consistent with the evidence once we combine four things:
 - outcomes are **right-skewed** — most earn little, a few earn enormously
 - founders collect large **non-pecuniary** benefits (autonomy, meaning)
 - many enter on **optimistic / overconfident** beliefs that overstate their odds
 - a skilled minority genuinely earns **high risk-adjusted returns** (ability + human capital)
- The puzzle is not “why does anyone do it?” — it is “**averaging hides everything that matters.**”

WHAT THIS MEANS FOR YOU

- Heritable **preferences and traits** (risk attitude, personality) tilt the odds of *entering* (recall the nature share in Lindquist et al. / Vladasel et al.) – but there is **no single “entrepreneur type.”**
- The things that most predict *doing well* – general and social ability, a balanced skill set, industry experience, a good team and network – are **buildable** (see also next class).
- Beware your own **optimism and overconfidence**: they help you start, but can also let you **persist with underperforming** projects and ideas.

SLIDES & DATA

The slides and the data behind every graph in this lecture are available at:

github.com/jmahieu/foundations-of-entrepreneurship